

```
lena=imread('Database\Images\Lena.jpg');  
lenasize=size(lena)
```

```
lenasize = 1×3  
512    512     3
```

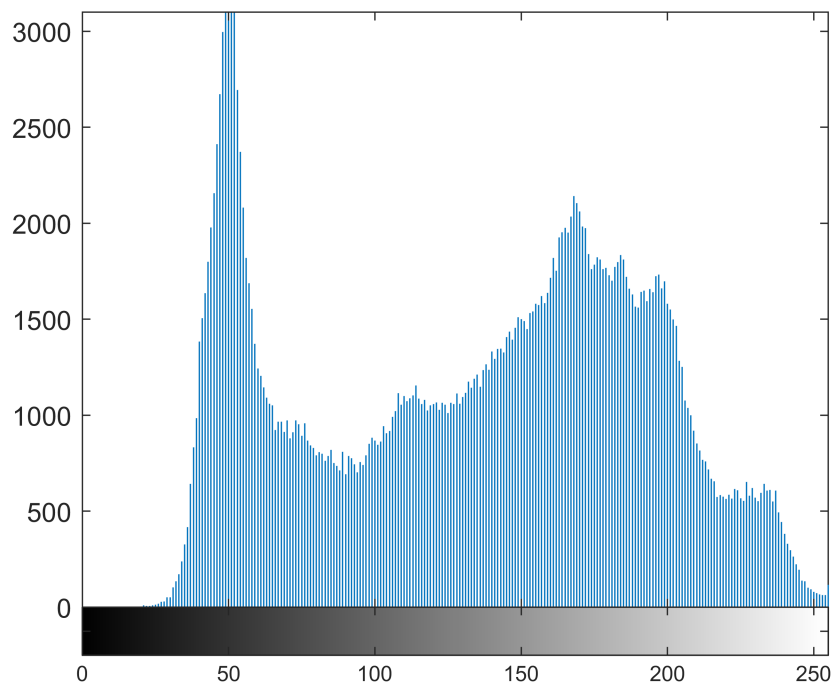
```
imshow(lena);
```



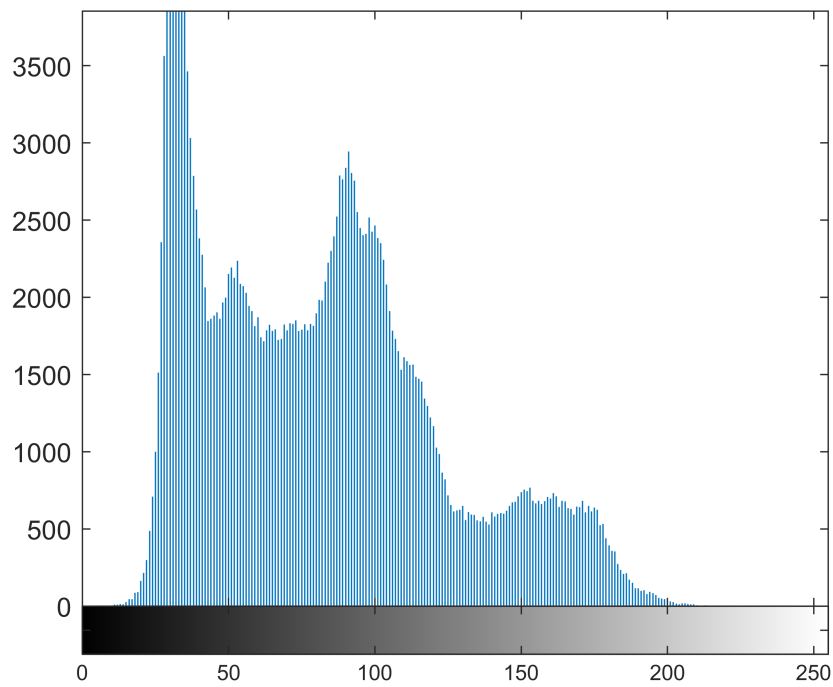
```
lenacut=lena(106:406,106:406,:);  
imshow(lenacut)
```



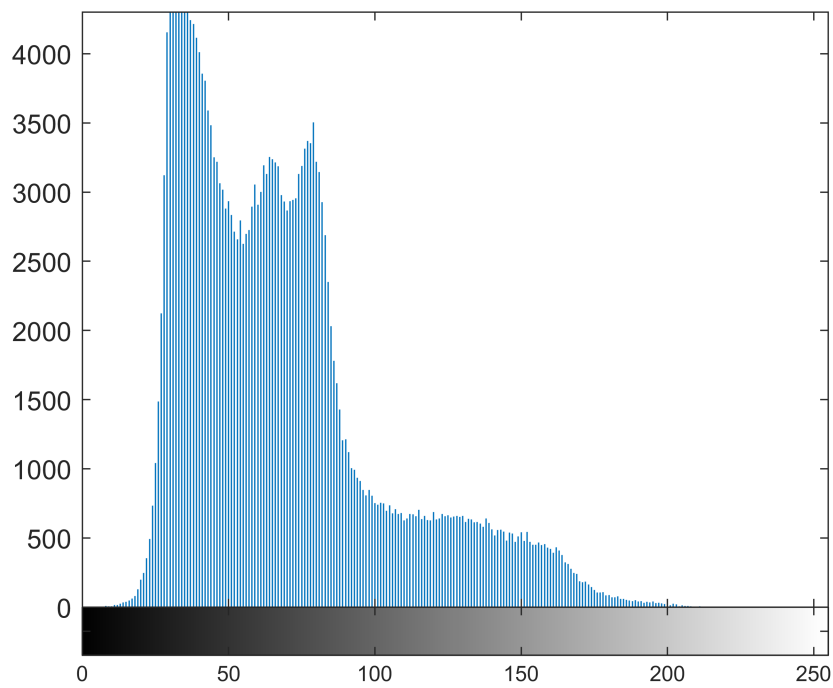
```
imhist(lena(:,:,1))
```



```
imhist(lena(:,:,2))
```



```
imhist(lena(:,:,3))
```



```
random1=uint8(rand(512,512).*255);  
random2=uint8((randn(512,512).*31)+127);  
  
max(max(random1))
```

```
ans = uint8
```

```
255
```

```
min(min(random1))
```

```
ans = uint8
```

```
0
```

```
mean(mean(random1))
```

```
ans = 127.1947
```

```
max(max(random2))
```

```
ans = uint8
```

```
255
```

```
min(min(random2))
```

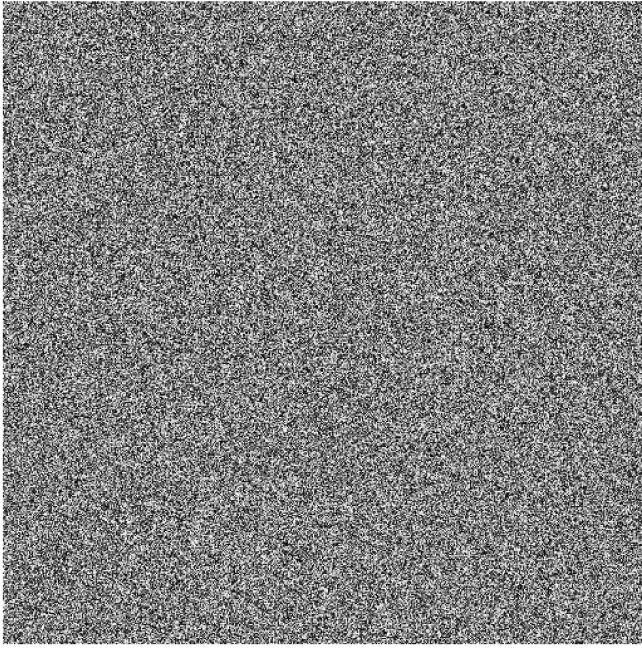
```
ans = uint8
```

```
0
```

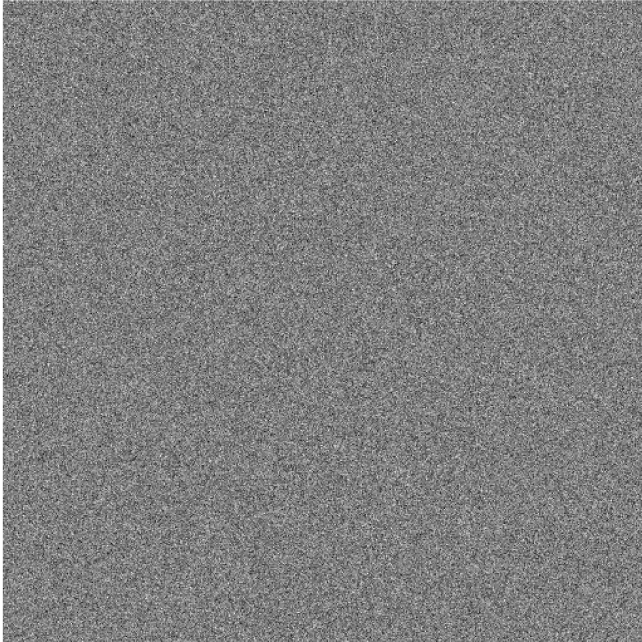
```
mean(mean(random2))
```

```
ans = 127.0172
```

```
imshow(random1)
```

```
imshow(random2)
```



```
lenagray = rgb2gray(lena);  
imshow(lenagray)
```

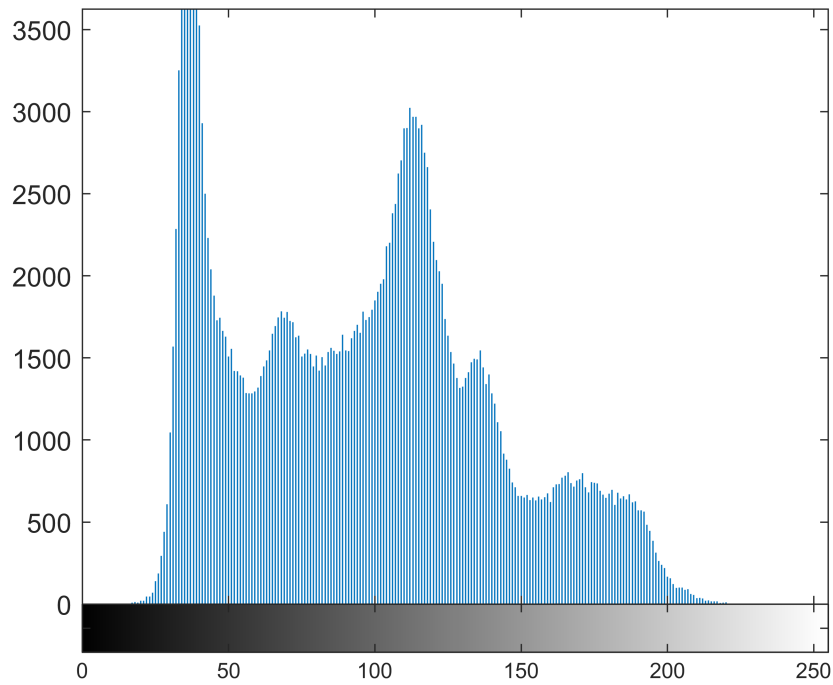


```
eqlenagrry=histeq(lenagray);  
imshow(eqlenagrry)
```

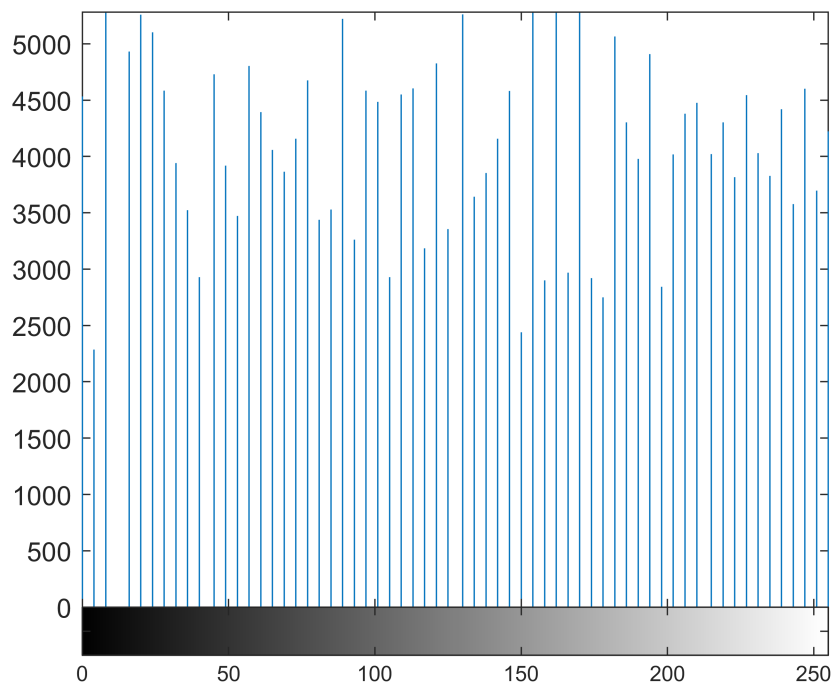


```
% subplot(2,1,1)
```

```
imhist(lenagray)
```



```
% subplot(2,1,2)  
imhist(eqlenagrry)
```



```

LP2=[1 2 1;2,4,2;1 2 1];
HP1=[-1 -2 -1;0 0 0;1 2 1];
HP2=[-1 0 1;-2 0 2;-1 0 1];
HP3=[0 -1 0;-1 4 -1;0 -1 0];

mask1=(1/9).*[1 1 1; 1 1 1; 1 1 1];
mask2=(1/sum(sum(LP2))).*LP2;

LP1f=uint8(conv2(lenagray,mask1));
LP2f=uint8(conv2(lenagray,mask2));
HP1f=uint8(conv2(lenagray,HP1));
HP2f=uint8(conv2(lenagray,HP2));
HP3f=uint8(conv2(lenagray,HP3));
HP4f=uint8(conv2(lenagray,HP3+[0 0 0;0 1 0;0 0 0]));

imshow(lenagray)

```



```

disp((1/9).*[1 1 1; 1 1 1; 1 1 1])

```

```

0.1111    0.1111    0.1111
0.1111    0.1111    0.1111
0.1111    0.1111    0.1111

```

```

imshow(LP1f)

```



```
disp((1/sum(sum(LP2))).*LP2)
```

0.0625	0.1250	0.0625
0.1250	0.2500	0.1250
0.0625	0.1250	0.0625

```
imshow(LP2f)
```



```
disp([-1 -2 -1;0 0 0;1 2 1])
```

```
-1  -2  -1  
0   0   0  
1   2   1
```

```
imshow(HP1f)
```



```
disp([-1 0 1;-2 0 2;-1 0 1])
```

```
-1    0    1  
-2    0    2  
-1    0    1
```

```
imshow(HP2f)
```



```
imshow(HP1f+HP2f)
```



```
disp([0 -1 0;-1 4 -1;0 -1 0])
```



```
0   -1   0
-1   4  -1
0   -1   0
```

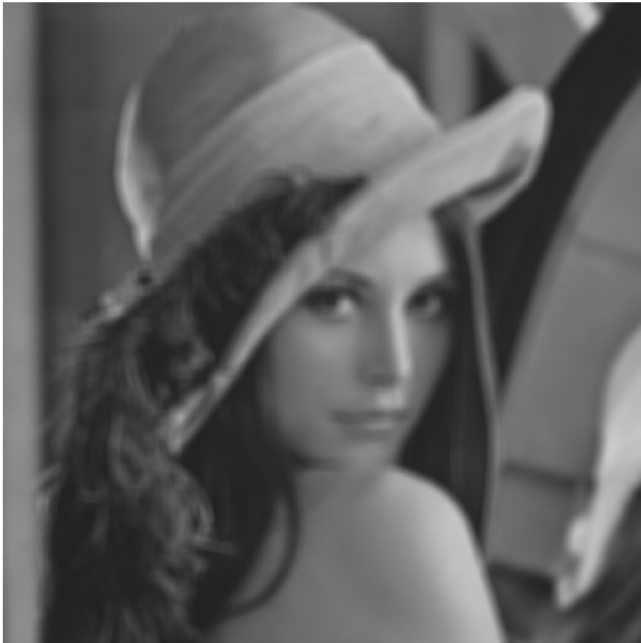
```
imshow(HP3f)
```



```
imshow(HP4f)
```



```
h = fspecial('average', [10 10]);  
im_filt = imfilter(lenagray,h,'replicate');  
imshow(uint8(im_filt));
```



```
h2 = fspecial('disk', 5);  
im_filt2 = imfilter(lenagray,h2,'replicate');  
imshow(uint8(im_filt2));
```



```
h3 = fspecial('gaussian', [10 10], 10);  
im_filt3 = imfilter(lenagray,h3,'replicate');  
imshow(uint8(im_filt3));
```



```
im_noise = imnoise(lenagray, 'salt & pepper', 0.02);  
im_filtered = medfilt2(im_noise);  
imshow(im_noise), figure, imshow(im_filtered)
```





```
h = fspecial('prewitt');  
im_filt = imfilter(lenagray,h,'replicate');  
imshow(uint8(im_filt));
```



```
h1 = fspecial('sobel');  
im_filt = imfilter(lenagray,h1,'replicate');  
imshow(uint8(im_filt));
```




```
h2 = fspecial('laplacian',1);  
im_filt = imfilter(lenagray,h2,'replicate');  
imshow(uint8(im_filt));
```



```
h3 = fspecial('log',[10 10],0.5);  
im_filt = imfilter(lenagray,h3,'replicate');  
imshow(uint8(im_filt));
```




```
h = fspecial('unsharp',1);  
im_filt = imfilter(lenagray,h,'replicate');  
imshow(uint8(im_filt));
```



```
imshow(edge(lenagray, "sobel", 0.04))
```



```
imshow(edge(lenagray, "canny", 0.04))
```



```
imshow(edge(lenagray,"roberts",0.04))
```

